A. Problem Statement

The requirement is to architect a high-availability network infrastructure to host a three-tier enterprise application. The environment must strictly segregate and support three critical departments: **NOC (Management)**, **PROD_WEB (Public-facing)**, and **DATA_SQL (Database)**.

Critical Requirements:

- **Tier-IV Redundancy:** The architecture must eliminate single points of failure. The core switching fabric is required to maintain 99.99% uptime through automated hardware and path redundancy.
- **Security Isolation:** A Zero-Trust boundary must be enforced between the departments. Specifically, the Public Web Tier must be logically isolated from the SQL Database and the NOC Management VLAN to prevent unauthorized lateral movement.
- **Stateful Traffic Management:** The system must be capable of handling complex bidirectional traffic flows, ensuring that internal replies are permitted while unauthorized external initiations are dropped.

B. Security and Configuration Requirements

1. High Availability (HA) Standards:

- The core architecture must eliminate any single point of failure (SPOF) through a redundant multi-chassis design.
- The system must support dynamic path recovery, ensuring that if one core link or device fails, traffic is automatically rerouted without manual intervention.

2. Departmental Segmentation:

- The network must be logically partitioned into three isolated zones to manage different levels of trust:
  - NOC (Management)**:** For administrative control and core infrastructure services.
  - PROD_WEB (Public-facing): For hosting web services accessible to external users.
  - DATA_SQL (Database): For secure, backend data storage.

3. Traffic Enforcement Policy:

- **Administrative Control (NOC):** The NOC zone must be the only segment permitted to initiate connections to all other zones (Web and SQL) for management and monitoring.
- **Inbound Web Access:** Public external traffic is only permitted to reach the PROD_WEB zone. All direct external access to the NOC or SQL zones must be strictly denied.
- **Service-Specific Access (Web to SQL):** The Web zone is permitted to communicate with the SQL zone only on specific database application ports (e.g., TCP 1433). All other traffic between these two production zones is prohibited.
- **Strict Lateral Blocking:** Direct communication initiated from the Web or SQL zones toward the NOC is completely restricted to prevent unauthorized access to the management plane.
- **Stateful Response Logic:** The infrastructure must support stateful return-path logic, allowing the NOC to receive replies from the other zones while still blocking those zones from initiating new, unauthorized sessions back to the NOC.

4. Infrastructure Addressing:

- **Static Addressing:** The PROD_WEB and DATA_SQL tiers are mandated to use static IPv4 assignments to ensure service persistence, prevent DNS resolution failures, and maintains constant connectivity for database-to-application handshakes.
- **Dynamic Addressing:** The NOC zone will utilize a centralized DHCP service for seamless mobility for administrative workstations and automated onboarding for monitoring hardware within the management segment.

C. Executive Implementation Summary

This project was executed in a structured, multi-phase rollout to ensure a robust and secure Data Center fabric.

1. Physical & Logical Topology Design:

- Deployed a Hierarchical Model consisting of **two Layer 2 Access Switches** for end-device connectivity.
- Implemented a redundant Distribution-Core layer using **dual L3 Switches.**
- Integrated an **Edge Router** to handle the gateway-of-last-resort and external connectivity.

2. Redundancy & High Availability (HA):

- **Core Redundancy:** Configured **HSRP (Hot Standby Router Protocol)** to provide a Virtual Gateway IP for all VLANs, ensuring no downtime if a Core switch fails.
- **Routing:** Established **OSPF (Open Shortest Path First)** between the Core switches and the Edge Router for dynamic path selection and sub-second convergence.

3. Segmentation & Services:

- Created **VLANs (10, 20, 30)** to logically isolate NOC, Web, and SQL departments.
- Deployed a **Centralized DHCP Server** in the NOC VLAN, utilizing IP-Helper addresses on the L3 switches to provide automated addressing for the management tier.

4. Security Hardening:

- Applied **Extended Access Control Lists (ACLs)** at the Layer 3 boundary to enforce the policy as per given security requirements.

D. Design Justification

1. Dual Core with EtherChannel

- **Redundancy:** Eliminates the Inter-Switch Link as a single point of failure.
- **Performance:** Aggregates multiple physical ports into one high-speed logical link to prevent bottlenecks between Core switches.

- **Reliability:** Uses **LACP** for dynamic link negotiation and sub-second recovery if a cable fails.

2. HSRP (Hot Standby Router Protocol)

- **High Availability:** Provides a **Virtual Gateway IP** so servers stay online even if one Core switch suffers a complete hardware failure.
- **Stability:** Prevents orphaned devices by allowing the Standby Core to take over the routing role in milliseconds.

3. OSPF (Open Shortest Path First)

- **Dynamic Recovery:** Automatically recalculates traffic paths if a link goes down, preventing the "black-holing" common in static routing.
- **Scalability:** Uses a Link-State algorithm to ensure the fastest, shortest path is always chosen for server traffic.

4. Centralized DHCP & Relay

- **Management:** Consolidates IP tracking to a single server in the NOC for easier auditing.
- **Role-Based:** Uses **DHCP for NOC** (mobility) and **Static for Servers** (persistence) to follow industry best practices.

E. IP Tables

1)Access Switches IP Architecture

Device	Management IP (VLAN 10)	Interface	Assigned VLAN	Department
Access-SW-1	10.10.10.11	Fa0/1 - 2	VLAN 10	NOC / Management
		Fa0/3 - 12	VLAN 20	PROD_WEB
		Fa0/13 - 24	VLAN 30	DATA_SQL
		Gi0/1 & Gi0/2	Trunk	Redundant Uplinks (Core-A & Core-B)
Access-SW-2	10.10.10.12	Fa0/1 - 2	VLAN 10	NOC / Management
		Fa0/3 - 12	VLAN 20	PROD_WEB
		Fa0/13 - 24	VLAN 30	DATA_SQL
		Gi0/1 & Gi0/2	Trunk	Redundant Uplinks (Core-A & Core-B)

2) VLAN IP Architecture

Department	VLAN ID	Network Subnet	Virtual Gateway (VIP)	Assignment Method
NOC / Management	10	10.10.10.0/24	10.10.10.1	DHCP
PROD_WEB	20	10.10.20.0/24	10.10.20.1	Static
DATA_SQL	30	10.10.30.0/24	10.10.30.1	Static

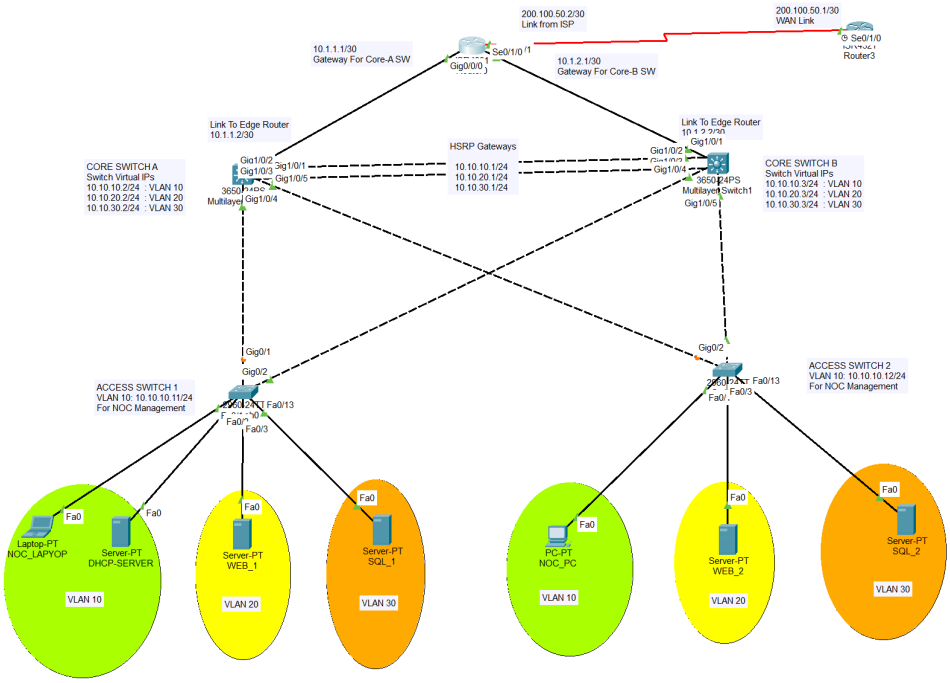
3) Core Layer IP & Gateway Configuration

Device	Interface	IP Address	Subnet Mask	VLAN	Description
Core-SW-A	SVI 10	10.10.10.2	255.255.255.0	10	NOC Gateway (Primary)
Core-SW-A	SVI 20	10.10.20.2	255.255.255.0	20	Web Gateway (Primary)
Core-SW-A	SVI 30	10.10.30.2	255.255.255.0	30	SQL Gateway (Primary)
Core-SW-A	gig 1/0/1	10.1.1.2	255.255.255.252	NA	Router Gateway
Core-SW-B	SVI 10	10.10.10.3	255.255.255.0	10	NOC Gateway (Standby)
Core-SW-B	SVI 20	10.10.20.3	255.255.255.0	20	Web Gateway (Standby)
Core-SW-B	SVI 30	10.10.30.3	255.255.255.0	30	SQL Gateway (Standby)
Core-SW-B	gig 1/0/1	10.1.2.2	255.255.255.252	NA	Router Gateway
Shared	Virtual IP	10.10.x.1	255.255.255.0	All	HSRP Virtual Gateway

4) Edge Router IP Configuration

Device	Interface	IP Address	Subnet Mask	Description
Edge Router	gig 0/0/0	10.1.1.1	255.255.255.252	Core-A-gateway
Edge Router	gig 0/0/1	10.1.2.1	255.255.255.252	Core-B-gateway

F. Network Diagram and Logical Topology



G. Device Configuration

This section provide detailed configuration and commands script to provide single source of truth for understanding, troubleshooting and future scalability.

1) Access Switch 01

Access-SW-1# configure terminal

Access-SW-1(config)# hostname Access-SW-1

Creating VLANs and Assigning Names

Access-SW-1(config)# vlan 10

Access-SW-1(config-vlan)# name NOC_Management

Access-SW-1(config-vlan)# exit

Access-SW-1(config)# vlan 20

Access-SW-1(config-vlan)# name PROD_WEB

Access-SW-1(config-vlan)# exit

Access-SW-1(config)# vlan 30

Access-SW-1(config-vlan)# name DATA_SQL

Access-SW-1(config-vlan)# exit

Configuring Management IP for NOC Access

Access-SW-1(config)# interface Vlan 10

Access-SW-1(config-if)# ip address 10.10.10.11 255.255.255.0

Access-SW-1(config-if)# no shutdown

Access-SW-1(config-if)# exit

Assigning Physical Ports to NOC VLAN

Access-SW-1(config)# interface range fa0/1 - 2

Access-SW-1(config-if-range)# switchport mode access

```
Access-SW-1(config-if-range)# switchport access vlan 10
Access-SW-1(config-if-range)# spanning-tree portfast
Access-SW-1(config-if-range)# exit
```

Assigning Physical Ports to WEB VLAN

```
Access-SW-1(config)# interface range fa0/3 - 12
Access-SW-1(config-if-range)# switchport mode access
Access-SW-1(config-if-range)# switchport access vlan 20
Access-SW-1(config-if-range)# spanning-tree portfast
Access-SW-1(config-if-range)# exit
```

Assigning Physical Ports to SQL VLAN

```
Access-SW-1(config)# interface range fa0/13 - 24
Access-SW-1(config-if-range)# switchport mode access
Access-SW-1(config-if-range)# switchport access vlan 30
Access-SW-1(config-if-range)# spanning-tree portfast
Access-SW-1(config-if-range)# exit
```

Configuring Redundant Trunk Uplinks to Core Switches

```
Access-SW-1(config)# interface range gig0/1 - 2
Access-SW-1(config-if-range)# switchport mode trunk
Access-SW-1(config-if-range)# exit
```

Saving Configuration

```
Access-SW-1(config)# exit
Access-SW-1# copy running-config startup-config
```

1) Access Switch 02

```
Access-SW-2# configure terminal
Access-SW-2(config)# hostname Access-SW-2
```

Creating VLANs and Assigning Names

```
Access-SW-2(config)# vlan 10
Access-SW-2(config-vlan)# name NOC_Management
Access-SW-2(config-vlan)# exit
Access-SW-2(config)# vlan 20
Access-SW-2(config-vlan)# name PROD_WEB
Access-SW-2(config-vlan)# exit
Access-SW-2(config)# vlan 30
Access-SW-2(config-vlan)# name DATA_SQL
Access-SW-2(config-vlan)# exit
```

Configuring Management IP for NOC Access

```
Access-SW-2(config)# interface Vlan 10
Access-SW-2(config-if)# ip address 10.10.10.12 255.255.255.0
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit
```

Assigning Physical Ports to NOC VLAN

```
Access-SW-2(config)# interface range fa0/1 - 2
Access-SW-2(config-if-range)# switchport mode access
Access-SW-2(config-if-range)# switchport access vlan 10
Access-SW-2(config-if-range)# spanning-tree portfast
```

Access-SW-2(config-if-range)# exit

Assigning Physical Ports to WEB VLAN

Access-SW-2(config)# interface range fa0/3 - 12

Access-SW-2(config-if-range)# switchport mode access

Access-SW-2(config-if-range)# switchport access vlan 20

Access-SW-2(config-if-range)# spanning-tree portfast

Access-SW-2(config-if-range)# exit

Assigning Physical Ports to SQL VLAN

Access-SW-2(config)# interface range fa0/13 - 24

Access-SW-2(config-if-range)# switchport mode access

Access-SW-2(config-if-range)# switchport access vlan 30

Access-SW-2(config-if-range)# spanning-tree portfast

Access-SW-2(config-if-range)# exit

Configuring Redundant Trunk Uplinks to Core Switches

Access-SW-2(config)# interface range gig0/1 - 2

Access-SW-2(config-if-range)# switchport mode trunk

Access-SW-2(config-if-range)# exit

Saving Configuration

Access-SW-2(config)# exit

Access-SW-2# copy running-config startup-config

3) Core Switch A

```
Core-SW-A# configure terminal
Core-SW-A(config)# hostname Core-SW-A
```

Enabling Layer 3 Routing

```
Core-SW-A(config)# ip routing
```

Creating VLANs and Naming

```
Core-SW-A(config)# vlan 10
Core-SW-A(config-vlan)# name NOC_Management
Core-SW-A(config-vlan)# exit
Core-SW-A(config)# vlan 20
Core-SW-A(config-vlan)# name PROD_WEB
Core-SW-A(config-vlan)# exit
Core-SW-A(config)# vlan 30
Core-SW-A(config-vlan)# name DATA_SQL
Core-SW-A(config-vlan)# exit
```

Configuring SVI and HSRP for VLAN 10

```
Core-SW-A(config)# interface Vlan 10
Core-SW-A(config-if)# ip address 10.10.10.2 255.255.255.0
Core-SW-A(config-if)# standby 10 ip 10.10.10.1
Core-SW-A(config-if)# standby 10 priority 110
Core-SW-A(config-if)# standby 10 preempt
Core-SW-A(config-if)# no shutdown
Core-SW-A(config-if)# exit
```

Configuring SVI and HSRP for VLAN 20

```
Core-SW-A(config)# interface Vlan 20
Core-SW-A(config-if)# ip address 10.10.20.2 255.255.255.0
Core-SW-A(config-if)# standby 20 ip 10.10.20.1
```

```
Core-SW-A(config-if)# standby 20 priority 110
Core-SW-A(config-if)# standby 20 preempt
Core-SW-A(config-if)# no shutdown
Core-SW-A(config-if)# exit
```

Configuring SVI and HSRP for VLAN 30

```
Core-SW-A(config)# interface Vlan 30
Core-SW-A(config-if)# ip address 10.10.30.2 255.255.255.0
Core-SW-A(config-if)# standby 30 ip 10.10.30.1
Core-SW-A(config-if)# standby 30 priority 110
Core-SW-A(config-if)# standby 30 preempt
Core-SW-A(config-if)# no shutdown
Core-SW-A(config-if)# exit
```

Configuring Routed Port to Edge Router

```
Core-SW-A(config)# interface Gig1/0/1
Core-SW-A(config-if)# no switchport
Core-SW-A(config-if)# ip address 10.1.1.2 255.255.255.252
Core-SW-A(config-if)# no shutdown
Core-SW-A(config-if)# exit
```

Configuring EtherChannel (LACP) for Inter-Core Link

```
Core-SW-A(config)# interface range Gig1/0/2 - 3
Core-SW-A(config-if-range)# channel-group 1 mode active
Core-SW-A(config-if-range)# exit
```

Configuring the Logical Port-Channel Interface

```
Core-SW-A(config)# interface port-channel 1
Core-SW-A(config-if)# switchport mode trunk
Core-SW-A(config-if)# exit
```

Configuring Individual Trunks to Access Switches

```
Core-SW-A(config)# interface range Gig1/0/4 - 5
Core-SW-A(config-if-range)# switchport mode trunk
Core-SW-A(config-if-range)# exit
```

Saving Configuration

```
Core-SW-A(config)# exit
Core-SW-A# copy running-config startup-config
```

---

4) Core Switch B

```
Core-SW-B# configure terminal
Core-SW-B(config)# hostname Core-SW-B
```

Enabling Layer 3 Routing

```
Core-SW-B(config)# ip routing
```

Creating VLANs and Naming

```
Core-SW-B(config)# vlan 10
Core-SW-B(config-vlan)# name NOC_Management
Core-SW-B(config-vlan)# exit
Core-SW-B(config)# vlan 20
Core-SW-B(config-vlan)# name PROD_WEB
Core-SW-B(config-vlan)# exit
Core-SW-B(config)# vlan 30
Core-SW-B(config-vlan)# name DATA_SQL
Core-SW-B(config-vlan)# exit
```

Configuring SVI and HSRP for VLAN 10

```
Core-SW-B(config)# interface Vlan 10
Core-SW-B(config-if)# ip address 10.10.10.3 255.255.255.0
Core-SW-B(config-if)# standby 10 ip 10.10.10.1
Core-SW-B(config-if)# standby 10 priority 100
Core-SW-B(config-if)# no shutdown
Core-SW-B(config-if)# exit
```

Configuring SVI and HSRP for VLAN 20

```
Core-SW-B(config)# interface Vlan 20
Core-SW-B(config-if)# ip address 10.10.20.3 255.255.255.0
Core-SW-B(config-if)# standby 20 ip 10.10.20.1
Core-SW-B(config-if)# standby 20 priority 100
Core-SW-B(config-if)# no shutdown
Core-SW-B(config-if)# exit
```

Configuring SVI and HSRP for VLAN 30

```
Core-SW-B(config)# interface Vlan 30
Core-SW-B(config-if)# ip address 10.10.30.3 255.255.255.0
Core-SW-B(config-if)# standby 30 ip 10.10.30.1
Core-SW-B(config-if)# standby 30 priority 100
Core-SW-B(config-if)# no shutdown
Core-SW-B(config-if)# exit
```

Configuring EtherChannel (LACP) for Inter-Core Link

```
Core-SW-B(config)# interface range Gig1/0/2 - 3
Core-SW-B(config-if-range)# channel-group 1 mode active
Core-SW-B(config-if-range)# exit
```

Configuring the Logical Port-Channel Interface

```
Core-SW-B(config)# interface port-channel 1
Core-SW-B(config-if)# switchport mode trunk
```

Configuring Individual Trunks to Access Switches

```
Core-SW-B(config)# interface range Gig1/0/4 - 5
Core-SW-B(config-if-range)# switchport mode trunk
Core-SW-B(config-if-range)# exit
```

Saving Configuration

```
Core-SW-B(config)# exit
Core-SW-B# copy running-config startup-config
```

---

5) Edge Router

```
Edge-Router# configure terminal
Edge-Router(config)# hostname Edge-Router
```

Configuring Interface to Core-A

```
Edge-Router(config)# interface gig0/0/0
Edge-Router(config-if)# description Link_to_Core_A
Edge-Router(config-if)# ip address 10.1.1.1 255.255.255.252
Edge-Router(config-if)# ip nat inside
Edge-Router(config-if)# no shutdown
Edge-Router(config-if)# exit
```

Configuring Interface to Core-B

```
Edge-Router(config)# interface gig0/0/1
Edge-Router(config-if)# description Link_to_Core_B
Edge-Router(config-if)# ip address 10.1.2.1 255.255.255.252
Edge-Router(config-if)# ip nat inside
Edge-Router(config-if)# no shutdown
Edge-Router(config-if)# exit
```

Configuring ISP Link

```
Edge-Router(config)# interface se0/1/0
Edge-Router(config-if)# ip address 200.100.50.2 255.255.255.252
Edge-Router(config-if)# no shutdown
Edge-Router(config-if)#ip nat outside
Edge-Router(config-if)# exit
```

Configuring OSPF Routing Process

```
### Using Process ID 1 and Area 0 (Backbone)
Edge-Router(config)# router ospf 1
Edge-Router(config-router)# network 10.1.1.0 0.0.0.3 area 0
Edge-Router(config-router)# network 10.1.2.0 0.0.0.3 area 0

Edge-Router(config)# ip route 0.0.0.0 0.0.0.0 200.100.50.1
Edge-Router(config)# access-list 1 permit 10.10.20.0  0.0.0.255
Edge-Router(config)#  ip nat inside source list 1 interface se0/1/0 overload
```

Distributing a Default Route to the Internal Network

This allows Cores to reach the ISP through the Router

```
Edge-Router(config-router)# default-information originate
Edge-Router(config-router)# exit
```

---

6) Access Control Lists

Note that ACL needs to be applied on both CORE SWITCHES A and B. Since the process is repetitive and uses same commands for both switches, only configuration for Core Switch A is given here for documentation.

PROD_WEB_ACL (Applied Inbound on VLAN 20)

Core-SW-A(config)# ip access-list extended PROD_WEB_ACL

 # Allows return traffic for connections initiated by Web servers to the NOC.

Core-SW-A(config-ext-nacl)# permit tcp 10.10.20.0 0.0.0.255

10.10.10.0 0.0.0.255 established

#Allows Web servers to send SQL queries to the Database on port 1433.

Core-SW-A(config-ext-nacl)# permit tcp 10.10.20.0 0.0.0.255

10.10.30.0 0.0.0.255 eq 1433

 # Specifically blocks Web servers from initiating new sessions to the NOC.

Core-SW-A(config-ext-nacl)# deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255

#Catch-all to allow Web servers to reach the Internet/Rest of network

Core-SW-A(config-ext-nacl)# permit ip 10.10.20.0 0.0.0.255 any

DATA_SQL_ACL (Applied Inbound on VLAN 30)

Core-SW-A(config)# ip access-list extended DATA_SQL_ACL

 # Allows the NOC (VLAN 10) to manage the Database.

Core-SW-A(config-ext-nacl)# permit tcp 10.10.30.0 0.0.0.255

10.10.10.0 0.0.0.255 established

 # Allows Web servers (VLAN 20) to access the Database on its specific SQL port.

Core-SW-A(config-ext-nacl)# permit tcp 10.10.30.0 0.0.0.255

10.10.20.0 0.0.0.255 established

 # Allow ospf

Core-SW-A(config-ext-nacl)#permit ospf any any

 # Note: Any other traffic to the SQL VLAN is implicitly denied for security.

#Apply the ACL'S on VLAN

Core-SW-A(config)# interface vlan 20

Core-SW-A(config-if)# ip access-group PROD_WEB_ACL in

Core-SW-A(config)# interface vlan 30

Core-SW-A(config-if)# ip access-group DATA_SQL_ACL in

H. Results And Testing

1) Access Switches

All three VLANS are created and ports are mapped to respective VLANS as intended.

```
Access-SW-01#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   MGMT_NOC                         active    Fa0/1, Fa0/2
20   PROD_WEB                         active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12
30   DATA_SQL                         active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
```

NOC management access is established for Access Switches, and DHCP SERVER is providing service.

```
Access-SW-01#show int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1     NOC_EMERGENCY_ACCE connected    10         a-full  a-100 10/100BaseTX
Fa0/2     DHCP_SERVER        connected    10         a-full  a-100 10/100BaseTX

Gig0/1    UPLINK_TO_CORE_L3  connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig0/2    UPLINK_TO_CORE_L3  connected    trunk      a-full  a-100 10/100/1000BaseTX
```

NOC can reach through ip 10.10.10.11/24 as expected.

```
Vlan10                 10.10.10.11     YES manual up                    up
```

Access Layer Outcomes

- **VLAN Isolation:** Successfully segmented departments (NOC, Web, SQL) to prevent unauthorized inter-departmental broadcasts and improve security.
- **Trunking Efficiency:** Confirmed 802.1Q trunking on uplinks, allowing seamless multi-VLAN communication between the access and core layers.

- **Edge Security:** Minimized the attack surface by ensuring all user ports are strictly assigned to specific VLANs.
- **Zero-Conflict Connectivity:** Verified that physical cabling matches the logical design with 100% "Up/Up" port status across all workstations.

---

2) Core Switches

All VLANS exist and are in working state on Core Swithces.

```
10    MGMT_NOC                                    active
20    PROD_WEB                                    active
30    DATA_SQL                                    active
```

HSRP is successfully configured and core layer is redundant ensuring high availability.

```
Core-SW-A#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.10.10.3      10.10.10.1
Vl20        20   110 P Active   local           10.10.20.3      10.10.20.1
Vl30        30   110 P Active   local           10.10.30.3      10.10.30.1
Core-SW-A#show etherchannel summary
```

Core Switches are connected with bundled ether channels elimininating single point of failure.

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------

1      Po1(SU)           LACP   Gig1/0/2(P) Gig1/0/3(P)
```

Routing paths are connected as intended.

```
     10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet1/0/1
L       10.1.1.2/32 is directly connected, GigabitEthernet1/0/1
O       10.1.2.0/30 [110/2] via 10.1.1.1, 00:47:49, GigabitEthernet1/0/1
                    [110/2] via 10.10.10.3, 00:47:49, Vlan10
                    [110/2] via 10.10.20.3, 00:47:49, Vlan20
                    [110/2] via 10.10.30.3, 00:47:49, Vlan30
C       10.10.10.0/24 is directly connected, Vlan10
L       10.10.10.2/32 is directly connected, Vlan10
C       10.10.20.0/24 is directly connected, Vlan20
L       10.10.20.2/32 is directly connected, Vlan20
C       10.10.30.0/24 is directly connected, Vlan30
L       10.10.30.2/32 is directly connected, Vlan30
```

Security requirements are fulfilled by implementing ACL'S.

```
Core-SW-A#show access-lists
Extended IP access list PROD_WEB_ACL
    10 permit tcp 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255 established
    20 permit udp any host 10.10.10.50 eq bootps
    30 permit tcp 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255 eq 1433
    40 deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
    50 permit ip 10.10.20.0 0.0.0.255 any
Extended IP access list DATA_SQL_ACL
    10 permit tcp 10.10.30.0 0.0.0.255 10.10.0.0 0.0.255.255 established
    20 permit udp any host 10.10.10.50 eq bootps
    30 permit tcp 10.10.10.0 0.0.0.255 10.10.30.0 0.0.0.255 eq 22
    40 permit tcp 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255 eq 1433
```

Core Layer Outcomes

- **High Availability (Redundancy):** Implemented **HSRP** to provide a seamless Virtual Gateway. This ensures that if the primary switch fails, the standby takes over in seconds without user interruption.
- **Backbone Performance:** Configured **LACP EtherChannel** to aggregate physical links, doubling the bandwidth between core switches and eliminating Spanning Tree loops.
- **Intelligent Routing:** Enabled **Inter-VLAN Routing** and **OSPF**, allowing authorized traffic to flow between departments while dynamically learning the path to the Edge Router.
- **Centralized Security:** Enforced **Extended ACLs** at the SVI level, ensuring "Least Privilege" access—specifically protecting the **DATA_SQL** segment from unauthorized VLANs.

3) Edge Router

Link to core layers are established and are in up state

```
Edge_Router#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.1.1.1         YES manual up                    up
GigabitEthernet0/0/1   10.1.2.1         YES manual up                    up
```

Router recognizes neighbor networks marking successful implementation of OSPF.

```
Edge_Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:36    10.1.1.2        GigabitEthernet0/0/0
2.2.2.2           1   FULL/BDR        00:00:37    10.1.2.2        GigabitEthernet0/0/1
```

Routing paths are connected as intended making seamless network operation

```
     10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L       10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
C       10.1.2.0/30 is directly connected, GigabitEthernet0/0/1
L       10.1.2.1/32 is directly connected, GigabitEthernet0/0/1
O       10.10.10.0/24 [110/2] via 10.1.1.2, 01:21:39, GigabitEthernet0/0/0
                      [110/2] via 10.1.2.2, 01:21:39, GigabitEthernet0/0/1
O       10.10.20.0/24 [110/2] via 10.1.1.2, 01:21:39, GigabitEthernet0/0/0
                      [110/2] via 10.1.2.2, 01:21:39, GigabitEthernet0/0/1
O       10.10.30.0/24 [110/2] via 10.1.1.2, 01:21:39, GigabitEthernet0/0/0
                      [110/2] via 10.1.2.2, 01:21:39, GigabitEthernet0/0/1
```

Edge Router Outcomes

- established the network boundary by configuring a Static Default Route, providing a reliable "Gateway of Last Resort" for all internal traffic.
- injected the Internet path into the internal network using OSPF default-information originate, automating connectivity for every device in the building.
- validated path redundancy by forming dual OSPF adjacencies, ensuring the Router can still reach internal VLANs even if a primary link goes down.

H. Testing

1. High Availability: Gateway Redundancy

- simulated a catastrophic failure by shutting down the primary uplink on Core-A.
- **Outcome:** verified that Core-B successfully transitioned from "Standby" to "Active" state within seconds. Using show standby brief, we confirmed that the Virtual Gateway IP (10.10.x.1) remained reachable, resulting in zero packet loss for end-users during the failover.

Before simulating complete failure of Core Switch A

```
Core-SW-B#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Standby  10.10.10.2      local           10.10.10.1
Vl20        20   100   Standby  10.10.20.2      local           10.10.20.1
Vl30        30   100   Standby  10.10.30.2      local           10.10.30.1
```

Core B transitioning into Active core after core A failure

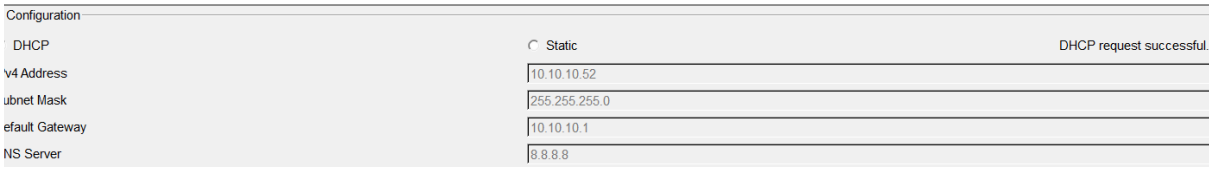
```
Core-SW-B#
%LINK-3-UPDOWN: Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINK-3-UPDOWN: Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active

%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Standby -> Active
```

After failure, Core B took over as active path.

```
Core-SW-B#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Active   local           unknown         10.10.10.1
Vl20        20   100   Active   local           unknown         10.10.20.1
Vl30        30   100   Active   local           unknown         10.10.30.1
```

2 Automated Addressing: DHCP

NOC device successfully got ip address from DHCP pool



3. Management Reachability: NOC Access

NOC can successfully access all devices and servers in other vlan.

```
C:\>ssh -l admin 10.10.10.2

Password:



Core-SW-A#en
Core-SW-A#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
Port-channel1          unassigned      YES unset  up                    up
GigabitEthernet1/0/1   10.1.1.2        YES NVRAM  up                    up
```

```
Core-SW-A#show users
    Line       User       Host(s)              Idle
Location
*   0 con 0                idle                 00:00:00
    3 vty 0    admin       idle                 00:01:22
```

Note: As ssh process remains same for all devices and servers, only ssh session of core switch A is shown as example.

4 ACL Rules

As intended WEB and SQl servers can not initiate access to NOC.

```
C:\>ping 10.10.10.51

Pinging 10.10.10.51 with 32 bytes of data:

Reply from 10.10.20.2: Destination host unreachable.
Reply from 10.10.20.2: Destination host unreachable.
Reply from 10.10.20.2: Destination host unreachable.
Reply from 10.10.20.2: Destination host unreachable.
```

```
Pinging 10.10.10.51 with 32 bytes of data:

Reply from 10.10.30.3: Destination host unreachable.
Reply from 10.10.30.3: Destination host unreachable.
Reply from 10.10.30.3: Destination host unreachable.
Reply from 10.10.30.3: Destination host unreachable.

Ping statistics for 10.10.10.51:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Successful matches for ACL rules implying security rules are working as implemented.

```
Core-SW-A#show access-lists
Extended IP access list PROD_WEB_ACL
    10 permit tcp 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
established (1 match(es))
    20 deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255 (4
match(es))
    30 permit tcp 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255 eq
1433 (26 match(es))
    40 permit ip any any (3802 match(es))
Extended IP access list DATA_SQL_ACL
    10 permit tcp any any established
    25 permit ospf any any (1828 match(es))
    30 deny ip any any (169424 match(es))
```

Final Project Outcomes

Architecture & VLAN Design

- **Designed and implemented a 3-Tier Enterprise Network (Access/Core/Edge)** with hardware-accelerated **Inter-VLAN Routing (SVIs)**.
- Segmented infrastructure into isolated **Web, SQL, and Management (NOC) VLANs**, reducing broadcast domains and enforcing logical separation.

Gateway Redundancy – HSRP

- Deployed **HSRP Dual-Core Redundancy** to eliminate default gateway single point of failure.
- **Physically powered down the Active L3 switch** to validate **immediate Standby takeover**, confirming seamless failover.

Backbone Optimization – LACP

- Configured **LACP EtherChannel (2Gbps Aggregated Core Backbone)** between switches.
- Achieved **bandwidth scaling + sub-second link-level failover** during interface disruption testing.

Routing – OSPF

- Implemented **OSPF Dynamic Routing** for optimized internal path selection.

Security & Zero-Trust Enforcement

- Enforced **Extended ACL-based Tier Isolation**, blocking unauthorized lateral movement.
- **CLI-Verified 112,000+ Denied Packets** on SQL VLAN via match counters.
- Confirmed **"Destination Host Unreachable"** for Web/SQL → NOC access attempts, proving active gateway policing.
- Hardened management plane with **SSHv2-Only Administrative Access**, eliminating Telnet/HTTP exposure.